

FINANCIAL CRISES

An Introduction

- Mridul Saggar
Professor of Practice (Economics)

Correspondence, if any: mksaggar@iimk.ac.in;

*This is for restricted use for the course participants alone & strictly not for circulation;
for appointments you can take the chance to walk in to my office
but appointment with a heads-up on mobile/WhatsApp or through Sathwik may be more beneficial*

What the Course is About?

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- It is all in the name! The course is about 'Financial Crises'.
- Financial is anything to do with Finance. Recall your Economic Environment (EE) course that served as macroeconomics 001.

Qs: What is finance? What is the difference between money and finance? Is money finance?

- *Crisis* was first used in a medical context, for the time in the when there is a major change in the intensity of disease that reflected a turning point of either recovery or death and by mid-17th century it too the figurative meaning of a "vitally important or decisive stage in the progress of anything", especially a period of uncertainty or difficulty. Today it reflects a period where stress (in your life, family, society, polity, nation or in the world) deepens to an extent where something needs to be done immediately to prevent a collapse. Crises is simply the plural of crisis.

Q: Is financial crisis same as economic crisis?



Dojima Rice Exchange,
Osaka, 1730 in Edo period

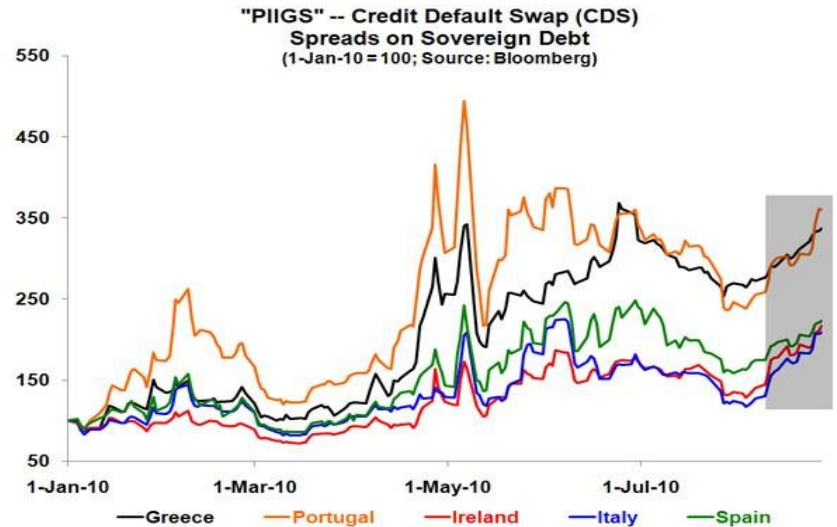


The rice exchange has
transformed itself to Osaka
Securities Exchange

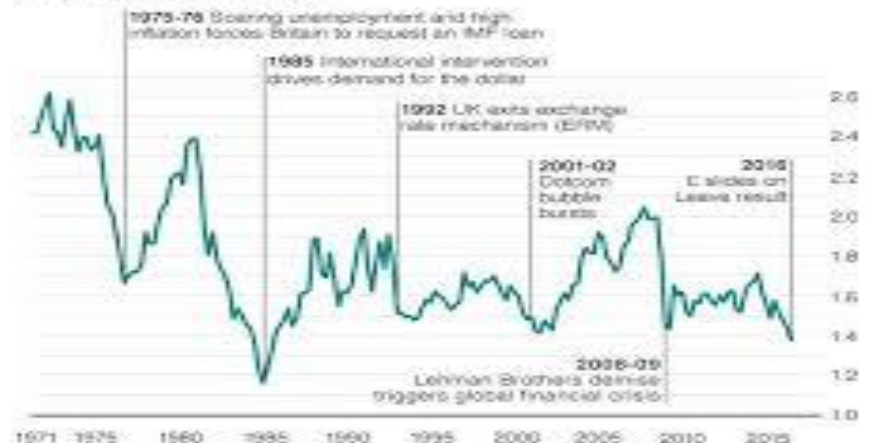
TYPES OF FINANCIAL CRISES

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- **The Sovereign Debt Crises**
 - *Driven by unsustainable public debt*
- **The Banking Crises**
 - *Driven by panics and runs on the bank*
- **The Currency Crises**
 - *Driven by fundamental misalignments of exchange rate, capital flow volatility and loss of reserves*



Value of sterling against US dollar since 1971
How many dollars £1 buys.



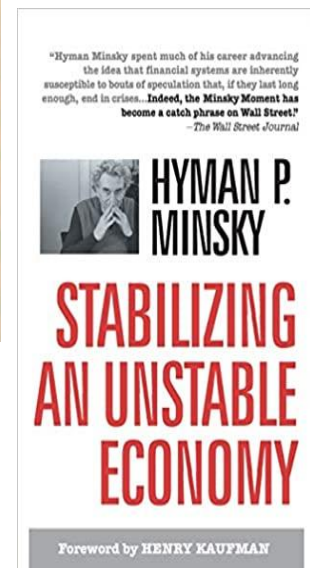
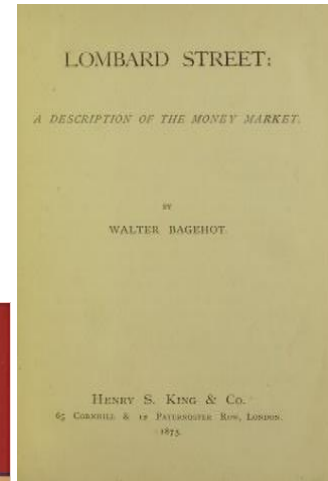
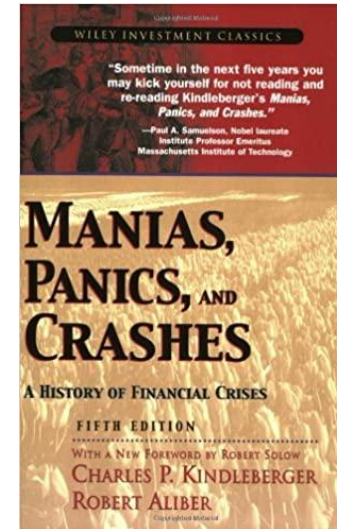
Note: Data as of 09:00 BST, 24 June 2016 (quarterly data)

Source: Bloomberg

KINDLEBERG-MINSKY MODEL OF FINANCIAL CRISES

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- **The Bagehot Rule (the basis of the LOLR function of the central banks):** In times of financial crisis central banks should lend **freely (i.e. liberally)** to **solvent** depository institutions, yet only against **sound collateral** and at interest **rate high enough (sort of penal rate)** to dissuade those borrowers that are not genuinely in need.
- **Charles P. Kindleberger:** Kindleberger analyzed hundreds of financial crises dating back centuries and found they share features of Hyman Minsky's model of the instability of a credit system.
- **Hyman P. Minsky:** Credit bubbles build on leverage as lending goes through three stages—hedge, speculative and ponzi and generates five stages of bubble: (i) displacement, (ii) boom; (iii) euphoria, (iv) profit taking and (v) panic



Course Facilitation

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ABOUT MYSELF (Mridul Saggar)

- Professor of Practice & Head of Uruppika: The Centre for Excellence in Macroeconomics, Banking & Finance (CEMBAF) [Joined IIMK: Dec 2022]
- Was a career central banker for three decades and retired from Reserve Bank of India (RBI) as its (i) Executive Director (in-charge of economic research and monetary policy); (ii) member of Monetary Policy Committee (MPC) that takes collegiate decision on policy repo rate and monetary policy stance and (iii) member of Financial Markets Committee that guided RBI interventions in money, G-sec & forex markets
- Was Head of the International Department of RBI during 2015-20 and was responsible for interactions with international financial institutions (IMF, World Bank, BIS, FSB), G20, SAARC and for relations with other central banks.
- Served as Chief Economist, Kotak Institutional Equities: 2008-10
- Served as Investor Education Protection Fund (IEPF) Chair Professor, NCAER during Jun-Nov 2022
- Served on IMF panel of monetary experts that advises its member countries on monetary policy and operating frameworks
- Appointed Chairman of the Technical Advisory Committee on Index of Industrial Production (TAC-IIP) by MosPI, GOI in October 2024

Email:

mksaggar@iimk.ac.in;

Mob: +919930099281

Faculty Block III, Room 15



ABOUT ACADEMIC
ASSOCIATE

(Sathwik Nayak)

Email: aaeco5@iimk.ac.in

Mob: +917019602246

Masters in Economics, Sri
Sahtya Sai Institute of Higher
Learning, Puttaparthi, AP

What all are the topics the course covers?

What do I need to read?



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- Please go through the course structure and gradually familiarize yourself with the course, structure, lecture topics, sub-topics
- There will be 24 lectures in all, of which ~first few are theoretical building blocks that includes a guest lecture on changing global economic order, next 4 are about international financial system and then lectures on case studies of some of the past major financial crises. The lectures about international financial system & on communication will be particularly helpful with preparations for the simulation of mock financial crisis exercise by learning about what international financial institutions do to mitigate, forestall and support handling of financial crises, including through bailouts. We will have some mock simulations from time to time but the great finale of financial crisis simulation will be in two back-to-back classes towards the end followed by a recap class. **[Please See Course Outline on Moodle]**
- Damn! The reading list looks vast and arduous. We are not mad to study journal articles/rigorous books in a one term course. We will put on the Moodle readings or their reference from time to time and you can try to learn as much as you can if you aim to outperform; but attending and focusing on the lectures and grasping the presentations handouts is a *sine qua non*.

How will I be graded?

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Assessment Tool	Weight	Remarks	PLO Tested (given the PLO No)
Quizzes of 15 objective type questions each	15%	Best of two out of three Quizzes (7.5% each) covering topics in the courses such as concepts on financial crises, history of financial crises, EWI and IFA and GFRR	PLO-1, PLO-2
Class Participation	5%	Including preparation for simulation exercise	PLO-2
Simulation Exercises (starting about Mid-term but spaced over in the second half of the course)	40% Individual assessment	Based on contribution to the group (team spirit) as well as individual performance and effectiveness of the role play, understanding of crisis-handling, leadership and oral & written communication	PLO-3, PLO-4
Final Exam	40%	Based on analytical and factual exam to assess understanding of the financial crisis and its handling	PLOs-1 to 4

Please check the Moodle for the Course Outline & the planned lectures & lecture schedule. The schedule can undergo minor change if needed, but you will be intimated.

Well, I am interested in fun part – the role play ... so, tell me how will we do the simulation exercise?

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- 1. Ministry of Finance Negotiation Group** (e.g. Finance Minister, Finance secretary and Chief Economic Advisor): Timeliness of approaching for IMF bailout; negotiations, especially on fiscal and structural adjustment components; drafting of Letter of Intent (LoI) and Memorandum of Economic & Financial Policies); media interactions; Chief Economic Advisor will be evaluated for their economic and financial analysis of the causes and remedies of crisis and internal memo preparations.
- 2. Central Bank Negotiation Group** (e.g. Governor, Deputy Governor, Chief Economist): Timeliness of approaching for IMF bailout; negotiations, especially on monetary and exchange rate components for central bankers; drafting of Letter of Intent (LoI) and Memorandum of Economic & Financial Policies); media interactions; Chief Economist will be evaluated for their economic and financial analysis of the causes and remedies of crisis and internal memo preparations.
- 3. IMF Team** (e.g. IMF Managing Director, IMF Mission Chief, IMF Regional Department Economist; Executive Director, IMF at country-office): Evaluation component will be based on drafting of Article IV Press Information Note (PIN), negotiation of conditionalities with the MOF & Central bank teams, media and stakeholders interactions.
- 4. Opposition Political Party:** Leader of the opposition, former Finance Minister, top labour Union leader
- 5. Media** (Editor of the Financial Daily; Financial Daily's correspondent assigned to track IMF loan negotiations): Evaluation component: preparation of press reports by correspondent on developments regarding IMF loan; interview by editor of Central Bank Governor; & of FM; interview of Mission Chief by correspondent
- 6. Debtors Negotiation Group** (e.g. foreign central bank having a swap arrangement with central bank of the country in distress, A major Multilateral/ Domestic Development Bank lender, foreign lender firm; domestic lender firm): Evaluation component: Each in debtor negotiation group will write a letter to IMF to protect debtor interest by making loan repayment a de facto or de jure condition for the loan. The haircut in repayment will need to be negotiated and will also be a key component of this part of evaluation.

I would be interested in this fun asap. So, can we please start learning about Financial Crises now? Yes, we can!

Why Are We Studying This Course?



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- Because you chose this elective! Because it seems to be a lot of fun! Yes, it is but it is all about learning with fun. It may also help you develop some **traits of a good business leader**.
- Yet, this will not be at the expense of theoretical rigour and empiricism. The course cuts across standard subjects and provides building blocks for understanding crisis-related concepts drawing from macroeconomics (monetary and fiscal policies), open economy macroeconomics and international finance (especially exchange rate economics and capital flows), financial regulation, international financial architecture (especially IMF, global financial regulatory reforms & international financial diplomacy).
- to understand the concepts of banking and currency crisis with real life simulated experience. It will not only provide you an understanding of financial crises but also of public policies to prevent, mitigate and deal with systemic financial crises.
- It is a sort of laboratory experiment of a real-life financial crises environment more as a learning tool and not a controlled experiment to produce research results. It is learning aid for open economy macro-economics and related theoretical concepts for the most part of the course.
- The course will better prepare you for jobs in the areas of financial markets, financial institutions, regulatory institutions, multilateral institutions, government and central banks
- It will help you in business and economic decision-making during a financial crisis and also develop your negotiation and writing skills.

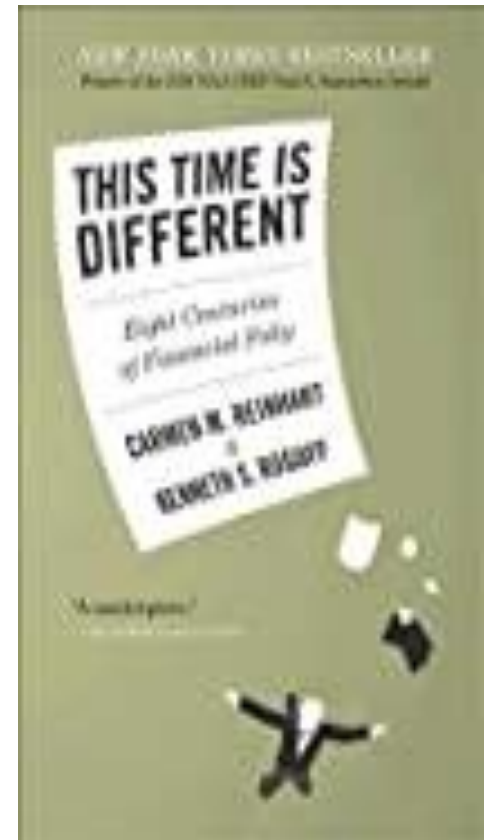
But can we now get into learning more about Financial Crisis? Well if you have no further questions about the course

IS EACH FINANCIAL CRISIS DIFFERENT FROM ANOTHER? ...(1)

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This time it is different –Eight Centuries of Financial Follies – Carmen M. Reinhart and Kenneth S. Rogoff (2011)

- The central bankers, policy makers and investors involved in every financial bubble are utterly convinced that, in terms of economic events, “this time is different.” They ignore the telltale signs of a bubble when they are in the grasp of “this-time-is-different syndrome.”
- Bankers and economists in the 1920s predicted that wars would not recur and the future would be stable.
- During 2003-07 central bankers and Wall Street justified soaring home prices and rising household debt, ignoring that they indicate a bubble.
- Currency debasement was common for centuries. In the past 100 years, inflation has replaced debasement.
- Sovereign defaults are a normal part of global capitalism, although they ebb and flow.
- Financial crises have occurred regularly over the past two centuries
- To avoid future bubbles, bankers and economists should use an early-warning system and a stricter regulatory scheme.



IS EACH FINANCIAL CRISIS DIFFERENT FROM ANOTHER? ...(2)

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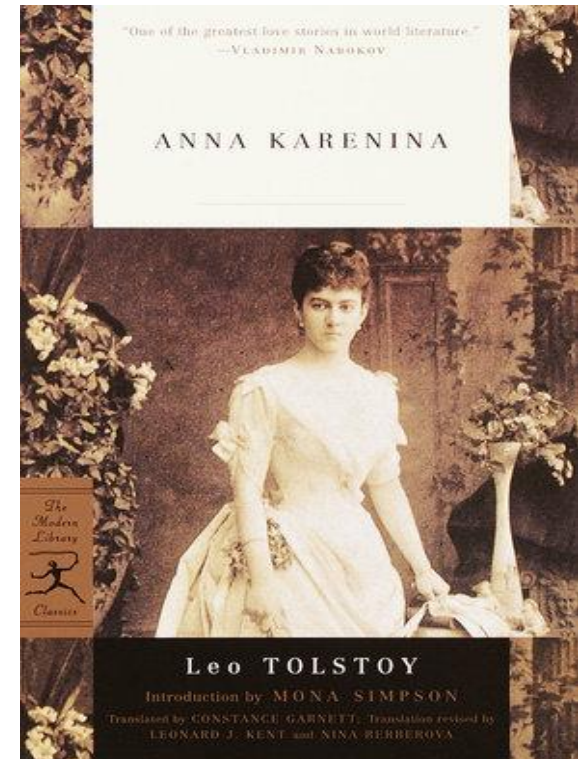
❑ The Anna Karenina Principle from Leo Tolstoy's 1877 novel:

All happy families are alike;

each unhappy family is unhappy in its own way

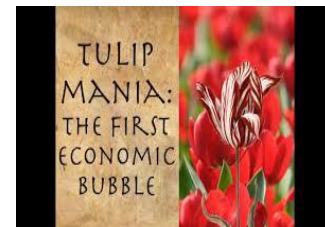
Common sets of attributes that lead to happiness:

- (i) sustainable growth around potential output
- (ii) Low and stable inflation
- (iii) A fiscal deficit that keeps debt sustainable
- (iv) A sustainable current account gap
- (v) Banks having adequate liquidity and capital buffers
- (vi) Corporate balance sheets that are not stressed ($D/E \leq 2$; $ICR \geq 1$)



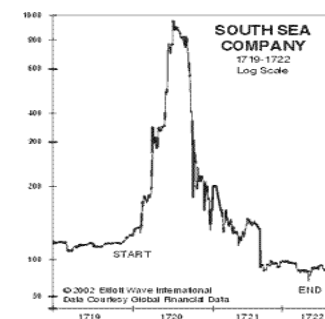
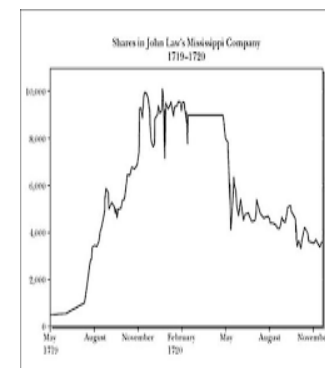
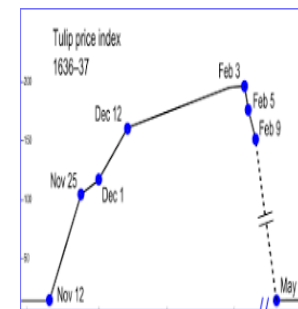
History is Replete with Financial Crisis

– Some early bubbles that crashed



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- **Tulip Mania (1634-38):** contract prices for some newly introduced fashionable Tulip bulbs in the formal futures market reached extraordinarily high levels and then dramatically collapsed in February 1637. At its height the Dutch craze was so much that some Tulip bulbs were sold at 5,500 guilders, the equivalent of a nice canal house in Amsterdam.
- **Mississippi Bubble (1718-20):** In 1716, John Law, an economic theorist born in Scotland, promised to revitalize a French economy ravaged by wars. He obtained permission to open a bank with the authority to issue notes and that would fund the government debt. A year later, the bank took over control of French trade in the Mississippi River valley. Law started to print banknotes to inflate the bank's stock price, but the scheme collapsed when note holders rushed to convert their notes into coin.
- **South Sea Bubble (1719-20):** In UK, South Sea Company, was founded in 1711 as a Public Private partnership with charter to trade and do fisheries in South Seas (now South Pacific), as also supply slaves. On June 11, 1720 when the British Parliament passed the Royal Exchange and London Assurance Corporation Act, popularly called the 'Bubble Act 1720', which forbade the formation of any other joint stock company unless approved by royal charter, this giving South Sea company virtually a monopoly power over its South Sea trade. the share price of the company that had been building up even before the Act as the company. In January 1720, the stock of the company was trading at £128 with total government debt at £50 million of which £18.3 million as held by three large corporations (South Sea company £11.7 mn; Bank of England £3.4 mn & East India Co: £3.2 mn) when decision was made to print £5 mn in money in face of England's war debt. South Sea company was printing notes in increasing quantity and its stock price initially zoomed up during Aug. 31 to Oct. 1, 1720 touching £750 but then collapsed to £175 by September.



History is Replete with Financial Crisis

– The Great Depression of the 1930s

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- **Great Depression (1929-33/ 1929-39):** It started with the Stock Market boom of 1920s that went bust in 1929 leaving deep scars on economy. The worst was over by 1933 but after effects continued till 1939. It has been the biggest, & the longest financial crisis ever that resulted in the deepest recession ever in the history
- Changes in major macro-parameters 1929-32:

	US	UK	France	Germany
Industrial production	–46%	–23%	–24%	–41%
Wholesale prices	–32%	–33%	–34%	–29%
Foreign trade	–70%	–60%	–54%	–61%
Unemployment	+607%	+129%	+214%	+232%



- US unemployment rate rose from 3.2% at the time of market crash to 24.9% in 1933; GDP contracted 8.5% in 1930 and 12.9% in 1932; inflation turned negative with deflation rates of 9.3% in 1931 and 10.3% in 1932

The Bretton Woods Conference, 1944

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- The Bretton Woods Conference, officially known as the United Nations Monetary and Financial Conference, was a gathering of delegates from 44 nations that met from July 1 to 22, 1944 in Bretton Woods, New Hampshire
- The Conference was called even while the World War-II was still going on to suggest a post-war global financial order
- The conference worked through three "commissions". Commission I worked on the IMF and was chaired by Harry Dexter White, Assistant to the Secretary of the U.S. Treasury and the chief American negotiator at the conference. Commission II worked on the IBRD and was chaired by John Maynard Keynes, economic adviser to the British Chancellor of the Exchequer and the chief British negotiator at the conference. Commission III worked on "other means of international financial cooperation" and was chaired by Eduardo Suárez, Mexico's Minister of Finance and the leader of the Mexican delegation. It was a venue for ideas that did not fall under the other two commissions
- The Conference agreed upon a series of new rules for the post-WWII international monetary system. They agreed to establish the International Monetary Fund (IMF) and the International Bank for Reconstruction and Development (IBRD) which is part of the present-day World Bank Group.



The Latin American Debt Crisis of the 1980s & the policy of heterodox shocks

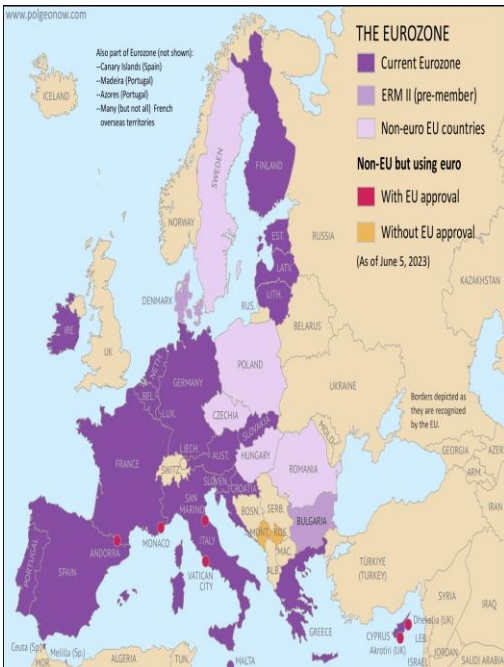
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- The routes of the crisis lay in the two large oil price shocks in the 1970s that caused widening of current account deficits in many Latin American (LatAm) countries. Oil-exporting countries, however, ran large current account surpluses that got parked in US banks and financial markets and then got on-lend to LatAm countries.
- total outstanding debt from all sources ballooned from only US \$29 billion (bn) in 1970 to US\$159 bn by the end of 1978 and , that number had skyrocketed to \$159 billion and further to US\$327 bn.
- In October 1979, Fed Chairman Paul Volcker announced new measures by the Federal Open Market Committee (FOMC) aimed at reining in the inflation that set off monetary policy tightening even across Europe and elsewhere.
- As US banks cut down their lending in response, the LatAm debt bubble burst. Mexico was the first to default, but many other LatAm countries followed.
- LatAm countries introduced policies of heterodox shocks – wage freezes, currency replacements in response to inertial inflation generated by high degree of indexation.



The European Exchange Rate Mechanism (ERM) Crisis

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Euro area as it is now

- When the ERM was set up in 1979, UK had refused to join. This was despite the Chancellor of Exchequer, Geoffrey Howe being a staunch European supporter. His successor Nigel Lawson was an advocate of fixed exchange rate system. But UK joined ERM only in 1990 and left in 1992 obtaining opt out from joining EMU in return of agreeing to sign Maastricht Treaty, which it did 1992 after securing two opt outs – (i) from euro and (ii) from Schengen.
- The 1992 sterling crisis peaked on “Black Wednesday” on 16 September 1992 when the UK government was forced to withdraw sterling from the ERM as it failed to keep its exchange rate above the lower limit required under the ERM. It was embarrassing as at that time, the UK held the Presidency of the Council of European Union (EU).
- In the run up to the Black Wednesday, George Soros through his Quantum Fund and some other currency traders had built huge short positions in sterling betting on the collapse of Sterling below the lower band of the ERM. On Black Wednesday he shorted about sterling £10 bn. Despite BoE making FX selling interventions that was said to have crossed £2 bn an hour at one stage and estimated to have cumulatively costed £22 bn in the run up to that Wednesday, sterling collapsed. BoE got defeated by Soros. Official estimates place the loss for BoE at £3.3 bn. Calculations showed that if BoE had maintained its £24 bn of FX reserves and sterling would have fallen the same amount as it did, BoE would have made profits of £2.4 bn. Bank of England has stopped intervening in FX markets after the debacle.

The Mexican Peso (Tequila) crisis of 1994-95

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Michael Camdessus, IMF MD



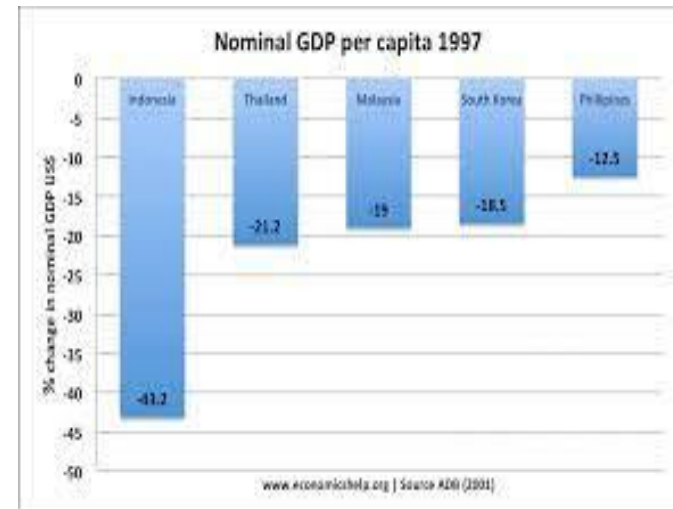
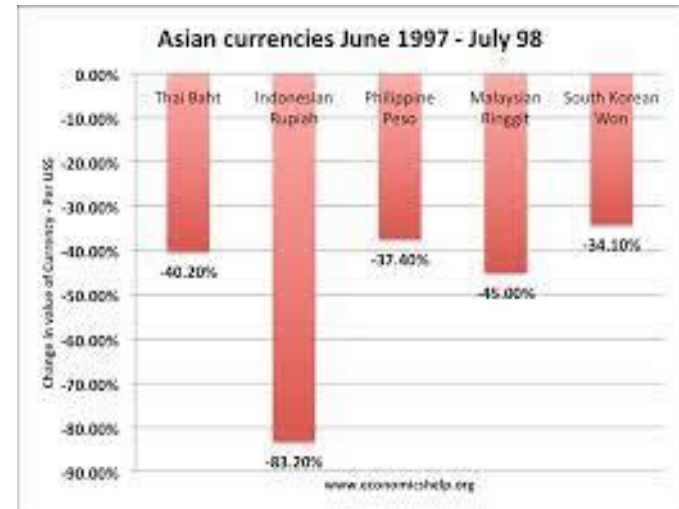
President Ernesto Zedillo

- The Mexican peso crisis was a currency crisis sparked by the Mexican government's sudden devaluation of the Mexican peso against US dollar by the new administration of President Ernesto Zedillo Ponce on 20th December 1994. The crisis was first of its kind ignited by capital flight (outflows) and was also the first case of market-led contagion amongst EM countries
- The crisis was generated by large dollar-indexed public debt borrowing by issuing Tesabonos when the going was good and investors wanted to put money in Mexico on the back of NAFTA getting formed. However, when the Fed suddenly tightened monetary policy raising Federal Funds rate, Mexico found it difficult to rollover debt and with depreciating Mexican peso found it difficult to pay its liabilities
- The contagion from the Mexican peso crisis affected neighbouring Latin American (LatAm) countries

Asian Financial Crisis (AFC)

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- The AFC was triggered with the collapse of Thai Baht peg with USD on July 2, 1997. With the Thai government floating Thai Baht capital flight ensued triggering chain across Southeast Asia and beyond. Not only the Korean won slumped but even the Japanese Yen, a hard currency, depreciated. Contagion spread to stock markets and other asset prices across the region and even in other parts of the world. Foreign debt to GDP ratio for the ASEAN-4 climbed from ~100% to ~180% as private debt defaults followed. South Korea, Thailand and Indonesia were worst affected, but Hong Kong, Malaysia and Philippines were also hurt. Contagions were felt into 1998.
- Indonesian President Suharto had to step down on May 21, 1998 on 21 May 1998 in the wake of widespread rioting that followed sharp price increases caused by a drastic devaluation of the [rupiah](#). The rupiah that traded in 2000-3000 range per US dollar in the early stages of crisis, fell to 16,800 in June 1998.
- Though the crisis unfolded as currency crisis, the roots lay in banking fragilities, especially in Thailand and Indonesia with high leverage and NPLs
- IMF lent about US\$118 bn in loans to Thailand, Indonesia and Korea to stabilize the region, while additional financial support came from Japan.



‘Taper Tantrum’ that hit financial markets in 2013?

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Taper Tantrum

[ˈtā-par ˈtan-trəm]

The 2013 surge in U.S. Treasury yields, resulting from the Federal Reserve's (Fed) announcement of future tapering of its policy of quantitative easing.

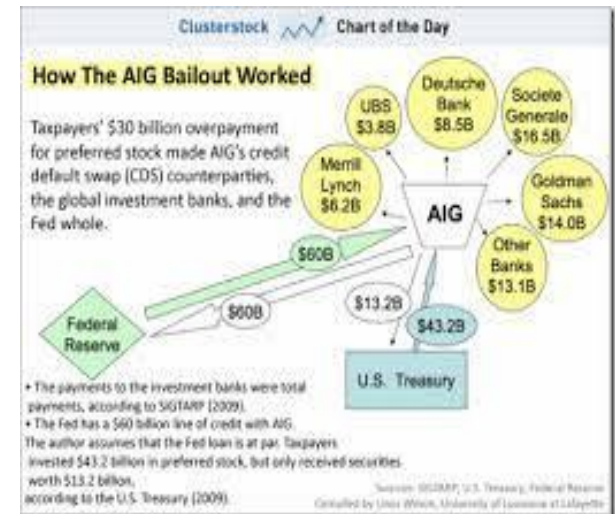


- Taper tantrum was an episode in 2013 that started with Fed Chair, ben Bernanke hinting at future tapering of its QE bond buying programme. US bond yields surged immediately and there were large spillovers to other countries, especially emerging markets which witnessed capital outflows.
- Starting with bond sell-off, the EM exchange rates faced depreciation pressures and then mayhem spread to equity markets as portfolio investors chose to sell-off equities as interest rate stocks were hit and then weakening EM currencies eroded foreign investor's profits. Derivative markets, especially futures, provided empathies to bears in the financial markets.
- It was not a full-scale financial crisis but a near one that could have turned into a prolonged one but for corrective central bank communications. There was a latent fear that the market would collapse if QE is withdrawn.
- Brazil, India, Indonesia, South Africa, Turkey became the fragile five that were worst impacted by market turmoil

Global Financial Crisis (GFC) 2007-09

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- The GFC was triggered by the sub-prime crisis in the US in 2007 and unfolded in 2008 with
- Sub-prime lending grew rapidly with poor lending standards to individuals (with FICO scores below 600).
- In Sept. 2007, Northern Rock was bailed out in UK but the problem lingered till it was nationalized in Feb. 2008; in March 2008 Bear Sterns was bailed out in US and with collapse of Lehman on Sept.15, 1988 GFC became a full-blown crisis across the world.
- Consequences of GFC were extreme. The net worth of American households had declined by about \$17 trillion in inflation-adjusted terms, a loss of 26%. In a 2018 study, the Federal Reserve Bank of San Francisco found that, 10 years after the start of the financial crisis, the country's GDP was approximately 7% lower than it would have been had the crisis not occurred, representing a loss of US\$70 thousand in lifetime income for every American. Approximately 7.5 million jobs were lost between 2007 and 2009, representing a doubling of the unemployment rate.



COVID-19 Pandemic Crisis of 2020-2021

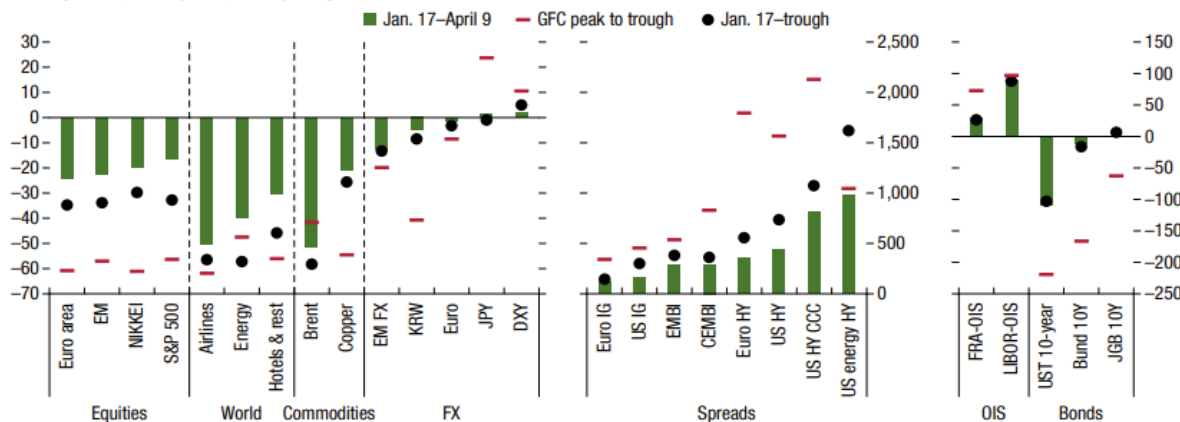
"Fear is an extraordinary jewel which has dominated human beings. If you can hold it and look at it, one begins to see the ending of it"

- Jiddu Krishnamurti at Brockwood Park 1984

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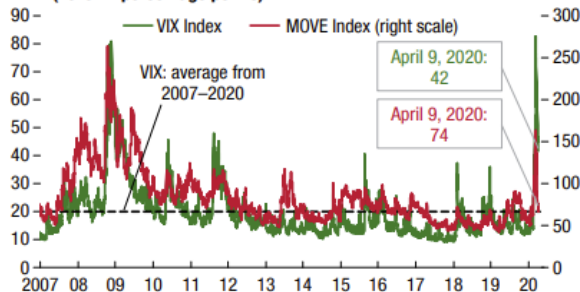
Investors fled risk assets for safe-haven assets, with some risk asset prices falling by more than 25 percent.

1. Asset Market Performance (as of April 9, 2020)
(Percent; basis points; basis points)



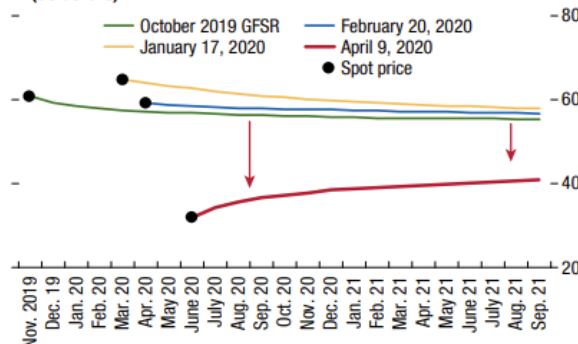
Market volatility spiked as COVID-19 spread globally.

2. Volatility Indexes
(Level in percentage points)



Oil prices collapsed as the OPEC+ deal fell apart on March 6.

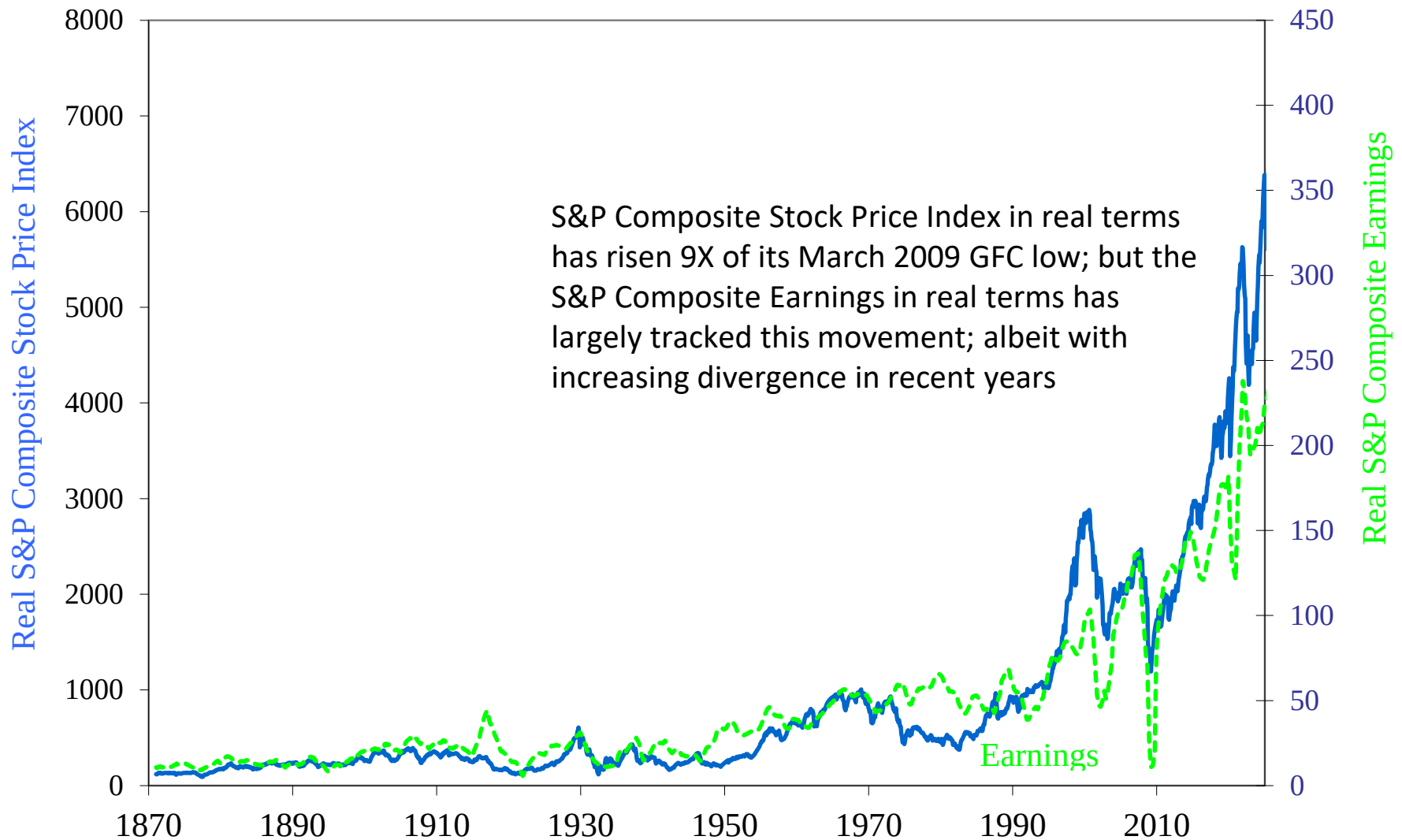
3. Oil Spot and Futures Prices
(US dollars)



- Asset prices tanked early on
- Some EM currencies dropped more than 20%
- Short-term funding markets came under stress
- Long-term borrowing costs soared as credit spreads widened
- Capital outflows from EMEs exceeded US\$100 billion in the first four months of CY; a scale not seen even at the time of GFC
- Financial conditions tightened abruptly
- Oil and non-oil commodity prices collapsed

Q: IS IT RIGHT TO THINK THIS TIME IT IS DIFFERENT AS BOOMING STOCK INDICES ARE ALSO TRACKED BY ACCELEAING COMPOSITE EARNINGS?

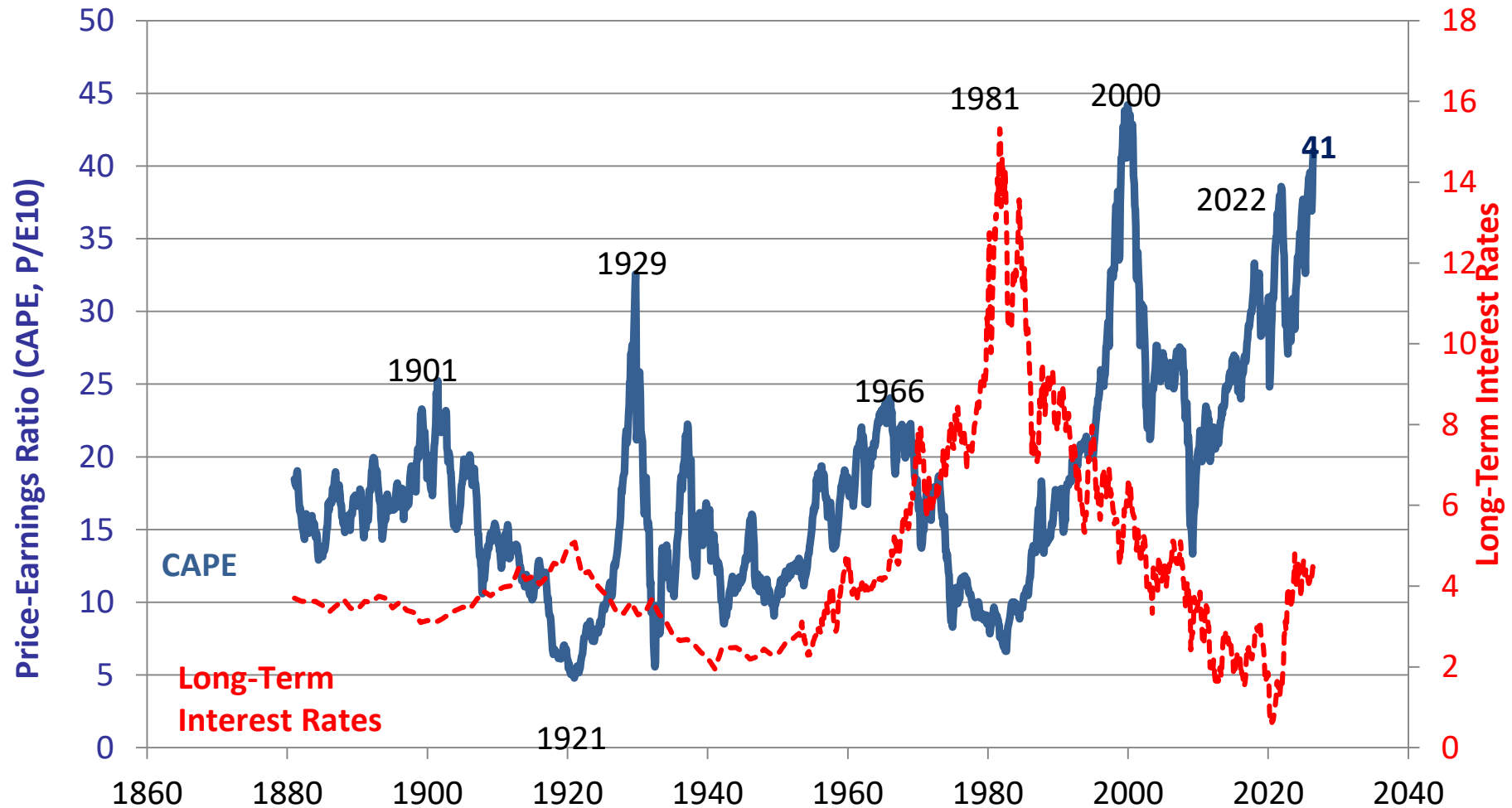
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CAPE currently at 41.0 is Nearing Historic Highs of 44.2

Q: ARE WE CLOSE TO EQUITY BUBBLE BURSTING CAUSING THE NEXT FINANCIAL CRISIS OCCURING?

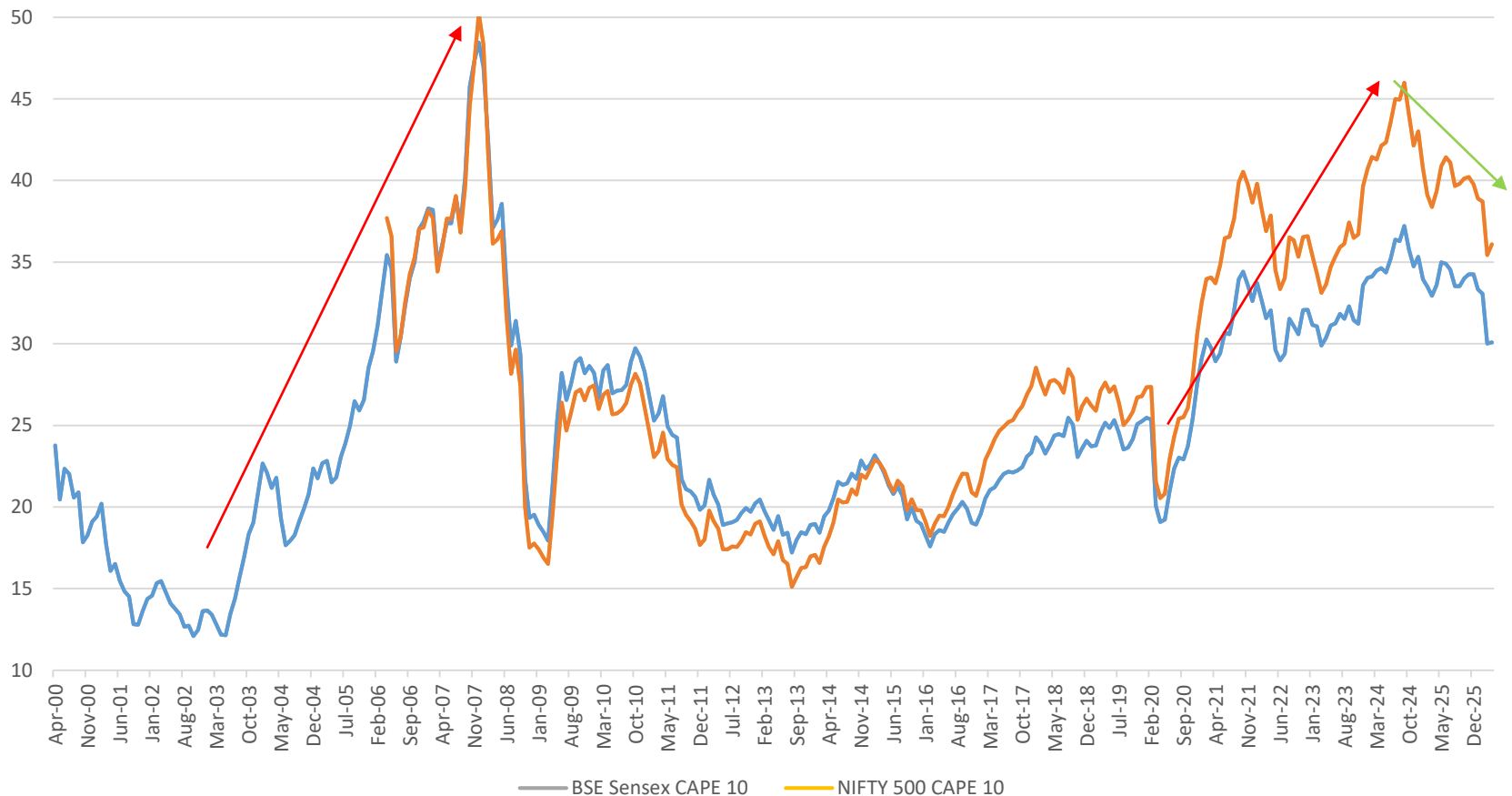
23



CAPE for Indian stock indices correcting after recent highs in Sept 2024 below the all-time highs in Dec-2007 during GFC

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SENSEX NIFTY CAPE10 Correcting



Q: IF THE BUBBLE BURSTS WHAT ASSET PRICES WILL BE AFFECTED & HOW? WILL THE CRASH BE LIKE Dotcom bubble burst; AFC; GFC OR THE PANDEMIC?

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